

NAME: SANIA SOHAIL

INTERN AT: Petalnex Pvt. Ltd.

INTERNSHIP DOMAIN: AI AUTOMATION

Day 31 — Requirement Analysis

1. Problem Statement

Recruiters receive job applications containing candidate information, experience, skills, and supporting material. Reviewing applications manually, checking completeness, assessing fit, recording outcomes, and sending candidate communication consumes time and can lead to inconsistent processing. This project proposes an n8n automation that validates incoming applications, uses AI for initial language-based screening, stores structured application information, routes the candidate to recruiter approval, updates the application status, and sends the appropriate email notification.

2. Current Process

Step	Current Process	Pain Point
1	Candidate submits application	Information may be incomplete or inconsistent.
2	Recruiter checks application	Manual validation is repetitive.
3	Recruiter reviews candidate fit	First-pass screening takes time.
4	Recruiter records outcome	Manual data entry can cause errors.
5	Recruiter communicates outcome	Candidate communication may be delayed.

3. Main Points

- Manual and repetitive application/resume review.
- Time spent extracting and normalizing candidate information.
- Inconsistent first-pass screening.
- Incomplete or incorrectly recorded candidate data.
- Delayed candidate communication.
- Lack of controlled fallback when AI processing fails.

4. Proposed Solution

- Trigger: Job Application Form receives a new application.
- Validation: Validate & Normalize checks required fields and standardizes input.
- Routing: Application Valid? separates valid applications from incomplete applications.
- AI: AI Screening evaluates application content against configured screening criteria.

- AI output: Parse AI Result converts the model response into structured data.
- Fallback: AI Fallback provides controlled handling when AI processing fails.
- Storage: Store Application persists application and screening information.
- Human review: Recruiter Approval captures the final recruiter decision.
- Decision: Approved? routes the recruiter outcome.
- Approved path: Mark Approved → Email Advance Candidate.
- Rejected path: Mark Rejected → Email Reject Candidate.
- Incomplete path: Email: Incomplete Application.

5. AI Role and Business Rules

- AI is used for language understanding and initial candidate screening.
- Validation, routing, status updates, and final approval are deterministic.
- AI output must be parsed/validated before downstream use.
- AI must not directly make the final hiring decision.
- Recruiter approval remains the final decision gate.

6. Suggested Structured AI Result

candidate_name, email, skills_match_score, experience_score, overall_score, recommendation (APPROVE/REJECT/REVIEW), reason, missing_requirements

7. Success Criteria

ID	Success Criterion	Target
SC-01	Valid applications follow the main path	Valid test submissions complete main path
SC-02	Missing required data is detected	Incomplete path is triggered
SC-03	AI produces structured output	Schema validates
SC-04	AI failure is controlled	Fallback/review path works
SC-05	Application is stored	Record exists with status
SC-06	Human approval is required	No automatic final hiring decision
SC-07	Correct email is sent	Email matches final status
SC-08	Secrets are protected	No credentials in exports/screenshots

8. Scope

- In scope: intake, validation, AI screening, parsing/fallback, storage, recruiter approval, status updates and email notifications.
- Out of scope: automatic employment decisions, payroll, background checks and unsupervised production hiring.

9. Expected Business Benefits

- Reduced repetitive recruiter administration.
- Faster first-pass processing.
- More consistent candidate records.
- Faster candidate communication.
- Clearer audit trail and human oversight.